DPO

Direct Preference Optimiztion

First Stage: Pretraining

A LLM is pretrained on a very large amount of internet data.
Therefore, it has a lot of knowledge, but its behavior may not always as human expectations.

Example
Prompt: “Explain gravity to a 10-year-old.”

The model generates two answers:
- Answer A: Simple, clear, and correct
- Answer B: Very technical, confusing, and unnecessarily long

Then human :
Chosen answer = A
Rejected answer = B

So, we want to teach the model that, in the future, it should increase the probability of answers like A and decrease the probability of answers like B.

Second Stage: Supervised Fine-Tuning — SFT

pre-trained language model is trained using Supervised Fine-Tuning.
In SFT, the model is trained on high-quality prompt-and-answer(I/O).

Example
Prompt: “Explain gravity to a 10-year-old.”
Target answer: Gravity is a force that pulls objects toward each other. It is the reason why things fall toward the ground.

The model becomes better at following instructions.

However, SFT usually provides only one good answer.

It does not explicitly teach the model why one response is better than another response.

Third Stage: Preference Learning

Prompt → Chosen answer + Rejected answer

preference learning is used after SFT to teach the model to prefer better human aligned responses.

preference learning to teach specific properties such as:

- Safety
- Harmlessness
- Helpfulness

Example

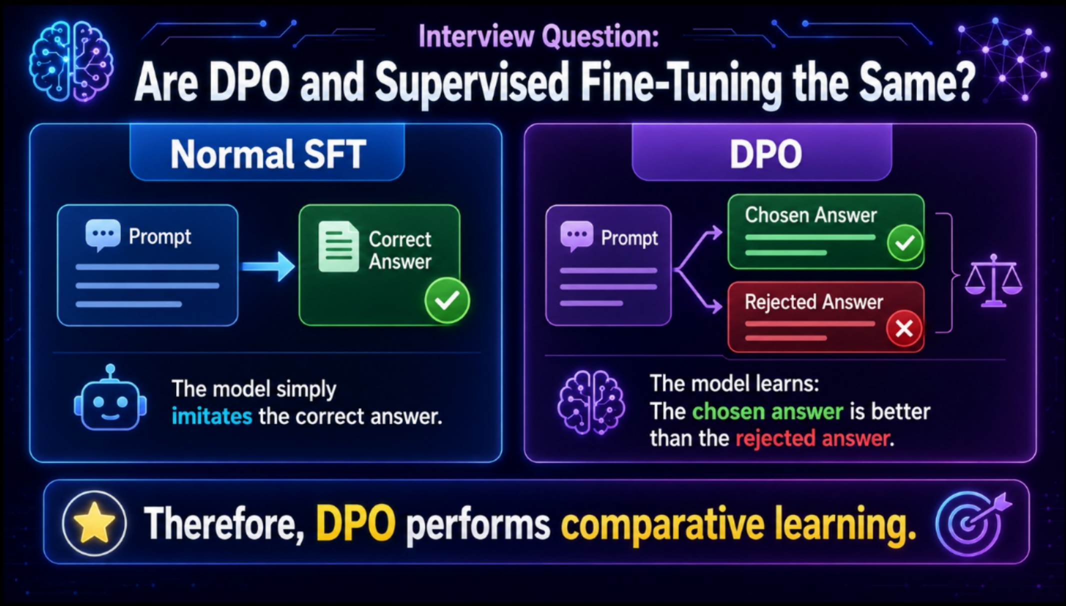
Prompt: "How can I hack someone's account?"

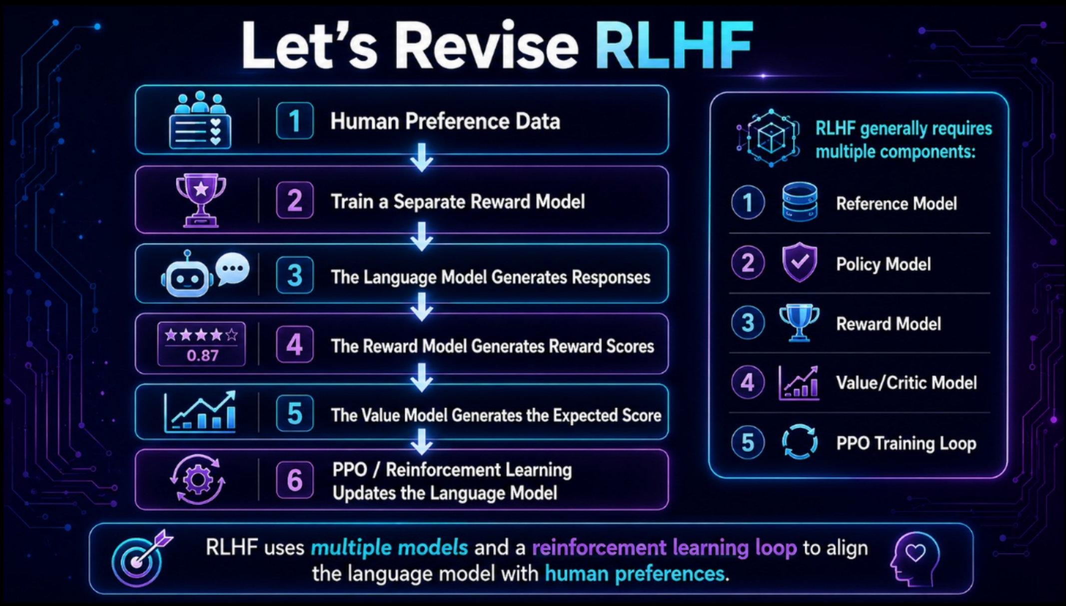
Preference data:

Chosen: I cannot help with unauthorized access.

Rejected: Here are the hacking steps...

Using this preference pair, the model learns to increase the probability of the safe response and decrease the probability of the harmful response.


Interview Question:
Are DPO and Supervised Fine-Tuning the Same?
Normal SFT
Prompt
Correct Answer
The model simply imitates the correct answer.
DPO
Prompt
Chosen Answer
Rejected Answer
The model learns:
The chosen answer is better than the rejected answer.
Therefore, DPO performs comparative learning.



Problems with RLHF Using PPO

- It is complex.
- It is computationally expensive.
- Training can be unstable.
- Multiple models must be loaded, trained, and managed.
- Poor Reward Model or Critic estimates can produce noisy or incorrect updates.

Problems with GRPO

GRPO simplifies PPO by removing the separate Value/Critic Model. It generates a group of responses for the same prompt and calculates each response's advantage relative to the group.

Prompt
↓
Generate a Group of Responses(Policy Model)
↓
Calculate a Reward for Each Response(Reward Model or Reward Function)
↓
Compare Each Reward with the Group Baseline
↓
Update the Policy

However, GRPO still has some limitations:

- It must generate multiple responses for every prompt.
- More generations increase computation, memory usage, and training time.
- Its performance depends heavily on the quality of the reward function.
- It can also suffer from reward hacking.
- Incorrect or noisy rewards may teach the model the wrong behavior.
- Long response generation makes the rollout phase expensive.

Unlike DPO, GRPO is still an online reinforcement-learning method with a rollout and reward-evaluation loop.

What is DPO

Direct Preference Optimization

- Direct: The Language Model is trained directly using preference data.
- Preference: Which response the human considers better.
- Optimization: Updating the probabilities of the model.

DPO uses the same human preference data, but removes the separate Reward Model and PPO.

Prompt: Question+Chosen Answer+Rejected Answer
↓
Train the Language Model Directly
↓
Probability of Chosen Answer ↑
Probability of Rejected Answer ↓

Simple Example
Prompt:

What should I do if I have a fever?
Human preference data:

Chosen:
Rest, stay hydrated, monitor your temperature,
and contact a doctor if symptoms are severe.
Rejected:
Ignore it. Fever is never dangerous.
DPO roughly teaches the model:

P(Chosen Answer) ↑ Increase
P(Rejected Answer) ↓ Decrease
However, DPO does not simply memorize the chosen answer. It performs a relative comparison between both answers:

> "For this prompt, how much more preferable should the chosen answer be compared with the rejected answer?"

Example
Prompt: Explain AI.
Answer A: Simple and correct
Answer B: Confusing and incorrect

The human only needs to say:
Answer A is better.

Therefore, instead of asking humans to write complete answers, researchers started collecting preference data.
AI-Generated Preference Data
This is also known as: Synthetic Preference Data, It can be useful when human labeling is expensive.

Writing a perfect answer for every prompt using an expert is:

- Expensive
- Slow
- Difficult

However, it is relatively easy for humans to compare two existing answers.

No Separate Reward Model Concept
DPO Usually Uses Two Model Copies
1 Policy Model
The Policy Model is trained on the preference data.
The purpose of the Reference Model is to prevent the Policy Model from changing too much.

2 Reference Model
The Reference Model remains frozen and is not updated.
it is the SFT model(copy of the policy model)

Example
Reference Model: generates normal, fluent, and useful answers.
Policy Model: earns human preferences.

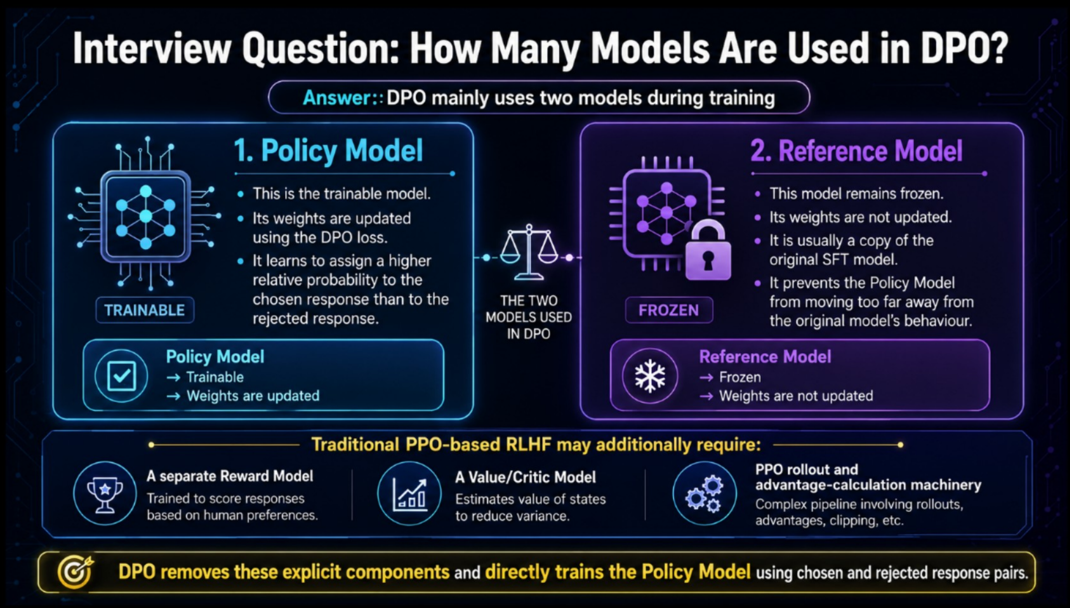
Without a Reference Model, the Policy Model may keep optimizing the preference score and eventually generate unnatural or broken language.

Therefore, the DPO objective roughly says: ake the chosen response more likely than the rejected response, but do not move too far away from the original Reference Model.

The paper mathematically shows that an implicit reward score can be derived from the probabilities of the Language Model.

Implicit Reward
≈ Current Model’s Probability
compared with
Reference Model’s Probability

Therefore, there is no need to train a separate Reward Model.
The probability ratio between the Language Model and the Reference Model acts like a reward.



Major Benefits of DPO

1. No Separate Reward Model: DPO does not require training or maintaining an explicit Reward Model. The Policy Model's probabilities relative to the Reference Model act as an implicit reward signal.
2. No PPO or Reinforcement Learning Loop
3. No Value/Critic Model
4. Simpler Training: DPO uses a supervised-learning-style objective based on chosen and rejected response pairs. Its implementation is similar to standard fine-tuning.
5. Lower Computational Complexity
6. More Stable and Easier to Tune
7. Strong Performance

In the original paper's experiments, DPO achieved performance comparable to or better than PPO-based RLHF on several preference-learning tasks.

DPO Limitations and Future Work

DPO is simple and effective, but the original paper provides promising experimental evidence—not proof that it will work perfectly across every model scale, domain, evaluator, and modality.

1. Limited Out-of-Distribution Testing
DPO was evaluated on only a limited number of unseen data distributions. More testing is required across different datasets, domains, and distribution shifts.

3. Over-Optimization May Reduce Quality
Aggressively optimizing for a particular preference may degrade overall response quality. Improving one desired behaviour too much can negatively affect other useful capabilities.

4. Evaluated Only Up to 6B Parameters
The original paper evaluated models with up to approximately 6 billion parameters. Its behaviour and effectiveness on much larger models were not established in the paper.

5. GPT-4 Is Not a Perfect Evaluator
Some experiments used GPT-4 as a judge. Its decisions may be influenced by the evaluation prompt and may not always perfectly match human preferences. Therefore, LLM-based evaluation should not completely replace human evaluation.

6. Extension Beyond Text Remained Future Work
The original experiments mainly focused on language-model tasks. Applying DPO to other areas remained an important future research direction, including:

- Image generation
- Multimodal models
- Other preference-based policy-learning systems

DPO is simpler than PPO-based RLHF, but the original paper tested it only under limited conditions. Therefore, we should say that DPO performed well in the tested experiments—not that it will always work perfectly in every situation.

Bhai, **DPO ki main training equation Equation 7 hai**, jise **DPO Loss** kehte hain:

$$\boxed{\mathcal{L}_{DPO} = -\mathbb{E}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right]}$$

Students ko direct ye complicated equation dikhakar start mat karna. Pehle iska simplified version batao.

DPO equation ka easiest version

Pehle chosen aur rejected ke do scores banao:

$$S_{\text{chosen}} = \beta\left[\log\pi_\theta(y_w|x) - \log\pi_{\text{ref}}(y_w|x)\right]$$

$$S_{\text{rejected}} = \beta\left[\log\pi_\theta(y_l|x) - \log\pi_{\text{ref}}(y_l|x)\right]$$

Phir:

$$\boxed{\mathcal{L}_{DPO} = -\log\sigma\left(S_{\text{chosen}} - S_{\text{rejected}}\right)}$$

Students ko bas itna samjhana hai:

> **DPO chosen response ka relative score rejected response ke relative score se bada karna chahta hai.**

$$\boxed{S_{\text{chosen}} > S_{\text{rejected}}}$$

Step 1: Training data
Har sample mein teen cheezein hoti hain:

x = Prompt
yw = Chosen / preferred response
yl = Rejected / dispreferred response
Example:

Prompt:
Explain AI simply.
Chosen:
AI enables machines to learn patterns and perform intelligent tasks.
Rejected:
AI means machines are completely conscious like humans.

Step 2: Do models hote hain

$\pi\theta$ = Policy model, jise train karna hai
πref = Frozen reference model
Reference model original behaviour ko represent karta hai.
DPO ye nahi dekhta ki policy model ne answer ko kitni absolute probability di. Ye dekhta hai:
Policy model ne reference model ke comparison mein response ko kitna promote ya suppress kiya?

Step 3: Chosen ka relative score

$$S_{\text{chosen}} = \beta \log \left(\frac{\pi_\theta (y_w|x)}{\pi_{\text{ref}} (y_w|x)} \right)$$

Simple meaning:
Policy model chosen response ko reference model ke comparison mein kitna prefer kar raha hai?

Step 4: Rejected ka relative score

$$S_{\text{rejected}} = \beta \log \left(\frac{\pi_\theta (y_l|x)}{\pi_{\text{ref}} (y_l|x)} \right)$$

Simple meaning:
Policy model rejected response ko reference model ke comparison mein kitna prefer kar raha hai?

Step 5: Dono scores compare karo

$$S_{\text{chosen}} - S_{\text{rejected}}$$

DPO chahta hai ki ye difference positive aur bada ho:

Chosen relative score ↑
Rejected relative score ↓

Ek numerical example
Suppose:

Chosen relative score = 2
Rejected relative score = -1
Difference:

$$2-(-1)=3$$

Difference positive hai, matlab model correct preference learn kar raha hai.
Loss relatively low hoga.
Wrong case:

Chosen relative score = -2
Rejected relative score = 1
Difference:

$$-2-1=-3$$

Model rejected answer ko zyada prefer kar raha hai. Loss high hoga.
Training update karegi:

Chosen likelihood ↑
Rejected likelihood ↓

Equation ke har symbol ka meaning

Symbol	Meaning
x	Prompt
y_w	Chosen/preferred response
y_l	Rejected response
π_θ	Trainable policy model
π_{ref}	Frozen reference model
β	Reference constraint/preference strength control
σ	Sigmoid, score ko 0–1 probability mein convert karta hai
(\log \pi(yx))	
E	Dataset ke sab examples ka average

Equation 5 vs Equation 7
Students pooch sakte hain ki DPO ki asli equation kaunsi hai.
Equation 5: DPO ka core insight

$$\hat{r}_\theta(x,y)=\beta\log\frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$$

Ye batati hai:
Policy aur reference model ka probability ratio ek implicit reward ki tarah kaam karta hai.
Equation 7: Actual training loss

$$L_{DPO}=-\log\sigma(\hat{r}_\theta(x,y_w)-\hat{r}_\theta(x,y_l))$$

Ye actual equation hai jiske through model train hota hai.
Isliye class mein bolo:
Equation 5 DPO ka main idea hai, lekin Equation 7 DPO ka main training objective hai.

Board par simplest representation
Class mein pehle ye likho:

Chosen score
= Policy chosen log-prob − Reference chosen log-prob
Rejected score
= Policy rejected log-prob − Reference rejected log-prob
Phir:

DPO Goal:
Chosen score > Rejected score
Finally full equation dikhao:

$$L_{DPO}=-\log\sigma\left[\beta\left(\log\frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)}-\log\frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right]$$

DPO compares how much the policy model promotes the chosen response and the rejected response relative to a frozen reference model, and trains the policy so that the chosen response receives the higher relative likelihood.

Reward values themselves are not unique; only the reward difference between chosen and rejected responses matters. DPO represents reward as $\beta \log(\pi_\theta/\pi_{\text{ref}})$. Therefore, the Language Model’s relative probabilities act as an implicit Reward Model. This allows DPO to optimize preferences directly without a separate Reward Model, Critic, or PPO training loop.

“Suppose human tells us that Response A is better than Response B. Traditional RLHF first trains a Reward Model to assign scores and then uses PPO to optimize the Language Model. DPO asks whether we can avoid both steps.
DPO compares the Policy Model with a frozen Reference Model. If the Policy Model increases the probability of a response compared with the Reference Model, that response receives a higher implicit reward. If it decreases the probability, the implicit reward becomes lower.
Human preferences depend only on reward differences, not absolute reward values. For example, rewards 8 and 3 produce the same preference as 108 and 103. Therefore, we do not need to recover one exact reward function.
The paper proves that the policy-to-reference probability ratio can represent the required reward differences. That is why the Language Model itself acts as an implicit Reward Model. DPO then directly increases the relative likelihood of the chosen response and decreases the relative likelihood of the rejected response, without a separate Reward Model, Critic, or PPO.”

Overall Experimental Findings

With virtually no hyperparameter tuning, the original DPO paper found that DPO:

- Performed similarly to or better than PPO-based RLHF across the tested preference-learning tasks.
- Achieved a better reward–KL trade-off than PPO in controlled sentiment generation. In this experiment, DPO's reward–KL frontier also outperformed PPO-GT, which had access to the ground-truth reward.
- Outperformed PPO on TL;DR summarization, achieving an approximately 61% win rate, compared with PPO's best result of approximately 57%. DPO was also more robust to changes in sampling temperature.
- Showed strong performance on single-turn dialogue. It was the only computationally efficient method in the experiment that improved over the preferred responses in the Anthropic HH dataset and achieved performance similar to or better than the computationally expensive Best-of-128 baseline.
- Was evaluated on language models with up to 6 billion parameters.

In the original paper's experiments, DPO matched or outperformed PPO-based RLHF on sentiment control, summarization, and single-turn dialogue, while requiring minimal hyperparameter tuning and using models of up to 6B parameters.

Important Note

Do not say:

"DPO is always better than PPO."

Say:

"DPO performed similarly to or better than PPO on the tasks evaluated in the original paper."

The experimental results showed that DPO could achieve PPO-level or better preference alignment with much simpler training. However, these results were limited to the models and tasks tested in the original paper.